

Chapter 4 - Classes of IP

1. What is an IP Address?

An **IP Address (Internet Protocol Address)** is a **logical address** used to uniquely identify a device on a network .

- It is called *logical* because it can be changed.
- It helps devices **find and communicate** with each other.

Daily Life Example:

- Just like your **home address** helps delivery people find your house ,
 - An IP address helps data find the correct device  .
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2. IP Address Size (IPv4)

- An IPv4 address is **32 bits** long
- These 32 bits are divided into **4 parts**
- Each part is **8 bits = 1 byte = 1 octet**

Example:

192.168.10.2

Here:

- 192 → 1st octet
 - 168 → 2nd octet
 - 10 → 3rd octet
 - 2 → 4th octet
-

3. What are Bits and Bytes?

◆ Bit

- Smallest unit of data
- Can be only **0** or **1** ☐ ☒

◆ Byte

- 8 bits = 1 byte

 1 Octet = 8 Bits = 1 Byte

Example:

- Switch ON = 1
- Switch OFF = 0

4. Binary Number System

- Binary works only with **0** and **1**
- Used by computers because hardware understands ON/OFF signals 

5. Powers of 2 (Very Important)

Each bit has a value from 2^7 to 2^0

Bit Position Power of 2 Value

2^7	128
2^6	64
2^5	32
2^4	16
2^3	8
2^2	4

Bit Position Power of 2 Value

2^1 2

2^0 1

👉 Total = 255 (Maximum value of 1 octet)

 6. Decimal to Binary Conversion (Step-by-Step)

 Rule:

- If number is needed → **ON (1)**
 - If not needed → **OFF (0)**
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 Example 1: Convert 192.168.10.2 to Binary

◆ **192**

$$128 + 64 = 192$$

128 64 32 16 8 4 2 1

1 1 0 0 0 0 0 0

Binary: **11000000**

◆ **168**

$$128 + 32 + 8 = 168$$

1 0 1 0 1 0 0 0

Binary: **10101000**

◆ **10**

$$8 + 2 = 10$$

0 0 0 0 1 0 1 0

Binary: **00001010**

◆ 2

0 0 0 0 0 0 1 0

Binary: **00000010**



Final Binary IP:

11000000.10101000.00001010.00000010



Example 2: Convert 172.191.121.24 to Binary

- 172 → 10101100
- 191 → 10111111
- 121 → 01111001
- 24 → 00011000



Final Binary:

10101100.10111111.01111001.00011000



Example 3: Convert 169.254.10.21 to Binary

- 169 → 10101001
- 254 → 11111110
- 10 → 00001010
- 21 → 00010101



Final Binary:

10101001.11111110.00001010.00010101

7. Types of IP Addresses

◆ IPv4

- 32 bits
- About **4.3 billion addresses**
- Written in decimal format

◆ IPv6

- 128 bits
- Uses **hexadecimal**
- Created because IPv4 addresses are finishing 🚨

8. Static vs Dynamic IP

📌 Static IP

- Fixed IP
- Used for servers 🖥️

📌 Dynamic IP

- Changes automatically
- Given by DHCP
- Used in homes 🏠

9. Private vs Public IP

🔒 Private IP

- Used inside office/home network
- Cannot access internet directly

Public IP

- Used on the internet
- Helps different networks communicate

Example:

- Office employees use private IP
 - Router uses public IP to access internet
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10. DNS Mapping

- DNS converts **domain names** → **IP addresses**

Example:

- google.com → 142.250.x.x

 Without DNS, we would need to remember IPs instead of names 

11. IP Address Classes

Class A

- Range: 0 – 127
- 1 Octet → Network
- 3 Octets → Host

Class B

- Range: 128 – 191
- 2 Octets → Network
- 2 Octets → Host
- About **65,000 hosts**

📌 Class C

- Range: 192 – 254
- 3 Octets → Network
- 1 Octet → Host
- Total = 256
- Usable hosts = **254**

👉 Why 254?

- 1 reserved for Network ID
 - 1 reserved for Broadcast ID
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📌 Class D

- Used for **Multicast** 📡

📌 Class E

- Reserved for **Research & Future use** 🧪
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🎯 Summary

- IP address = Logical address
 - IPv4 = 32 bits
 - Binary = 0 and 1
 - Decimal to Binary uses 2^7 to 2^0
 - Classes define network & host part
 - IP helps devices communicate
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CHAPTER SUMMARY



What is an IP Address?

An **IP Address (Internet Protocol Address)** is a **logical address** used to **uniquely identify a device** on a network 🌐.

◆ Called *logical* because:

- It **can be changed**
- Assigned by network or ISP



Purpose:

- Helps devices **locate** each other
- Enables **data communication**



Daily Life Example:

- Home address 🏠 → finds your house
- IP address 💻 → finds your device



IPv4 Address Size

- IPv4 = **32 bits**
- Divided into **4 parts**
- Each part:
 - 8 bits = 1 byte = 1 octet



Example:

192.168.10.2

Octet Value

1st 192

2nd 168

Octet Value

3rd 10

4th 2

Bits & Bytes

◆ Bit

- Smallest unit of data
- Can be **0 or 1** ☐ ☒

◆ Byte

- **8 bits = 1 byte**
- Also called **1 octet**

Rule:

1 Octet = 8 Bits = 1 Byte

Example:

- Switch ON → 1
- Switch OFF → 0

Binary Number System

- Computers use **binary (0 & 1)**
- Based on **ON / OFF electrical signals** ⚡

 Human → Decimal

 Computer → Binary

Powers of 2 (MOST IMPORTANT 🔥)

Each bit has a positional value:

Bit Position Power Value

2^7 128

2^6 64

2^5 32

2^4 16

2^3 8

2^2 4

2^1 2

2^0 1

👉 Maximum value of 1 octet:

$$128+64+32+16+8+4+2+1 = 255$$

Decimal to Binary Conversion

Rule:

- Needed value → **1**
- Not needed → **0**

Example 1: 192.168.10.2

192

$$128 + 64 = 192$$

Binary → 11000000

168

$$128 + 32 + 8$$

Binary → 10101000

10

8 + 2

Binary → 00001010

2

Binary → 00000010



Final Binary IP:

11000000.10101000.00001010.00000010

✓ **Example 2: 172.191.121.24**

172 → 10101100

191 → 10111111

121 → 01111001

24 → 00011000

✓ **Example 3: 169.254.10.21**

169 → 10101001

254 → 11111110

10 → 00001010

21 → 00010101

🌐 **Types of IP Addresses**

◆ **IPv4**

- 32 bits
- ~4.3 billion IPs
- Decimal format

◆ IPv6

- 128 bits
 - Hexadecimal format
 - Created due to **IPv4 exhaustion** 🚨
-

🔄 Static vs Dynamic IP

📌 Static IP

- Fixed IP
- Manually configured
- Used for **servers** 💻

📌 Dynamic IP

- Changes automatically
 - Assigned by **DHCP**
 - Used in homes 🏠
-

🏠 Private vs Public IP

🔒 Private IP

- Used inside LAN
- Not reachable directly from Internet

🌐 Public IP

- Used on Internet
- Provided by ISP

💡 Example:

- Laptop → Private IP
- Router → Public IP

DNS Mapping

- DNS converts:

Domain Name → IP Address

 Example:

google.com → 142.250.x.x

 Without DNS, internet is unusable 

IP Address Classes

Class A

- Range: **0 – 127**
- 1 Octet → Network
- 3 Octets → Host
- Very large networks

Class B

- Range: **128 – 191**
- 2 Octets → Network
- 2 Octets → Host
- ~65,000 hosts

Class C

- Range: **192 – 254**
- 3 Octets → Network
- 1 Octet → Host

📌 Total = 256

📌 Usable = **254**

? Why 254?

- 1 → Network ID
- 1 → Broadcast ID

📌 Class D

- Used for **Multicast** 📡

📌 Class E

- Research & future use 🧪

🎯 2 CONCLUSION

IP Addressing is the **core foundation of networking** 🌐

Without IP:

- ❌ No communication
- ❌ No Internet
- ❌ No Cyber Security

This chapter explains:

- ✓ How devices are identified
- ✓ How data finds destination
- ✓ How networks are structured

👉 **Every networking & cyber security role depends on IP knowledge** 📁 🔥



3 MIND MAP

Lecture 4 – IP Address

- |
- | — IP Address
 - | — Logical
 - | — Unique
 - | — Identifies device
- |
- | — IPv4 Structure
 - | — 32 bits
 - | — 4 octets
 - | — Max 255
- |
- | — Binary System
 - | — 0 & 1
 - | — Powers of 2
 - | — Decimal → Binary
- |
- | — Types
 - | — IPv4
 - | — IPv6
- |
- | — IP Nature
 - | — Static
 - | — Dynamic

| └— Private

| └— Public

|

└— DNS Mapping

| └— Name → IP

|

└— IP Classes

 └— Class A

 └— Class B

 └— Class C

 └— Class D

 └— Class E



Q & A

? Q1. What is an IP address?

✓ Answer:

An IP address is a logical address used to uniquely identify a device on a network.

? Q2. Why IP is called logical?

✓ Answer:

Because it can be changed and is assigned by the network.

? Q3. What is IPv4 size?

✓ Answer:

IPv4 is 32 bits long.

? Q4. How many octets in IPv4?

✓ Answer:

4 octets, each of 8 bits.

? Q5. What is a bit and byte?

✓ Answer:

Bit is smallest data unit (0/1).

8 bits = 1 byte.

? Q6. Why binary is used?

✓ Answer:

Because computers understand ON/OFF electrical signals.

? Q7. Max value of one octet?

✓ Answer:

255.

? Q8. Why not 256?

✓ Answer:

Because 8 bits give values from 0 to 255 only.

? Q9. Difference between IPv4 and IPv6?

✓ Answer:

IPv4 = 32 bit

IPv6 = 128 bit

? Q10. What is static IP?

✓ Answer:

A fixed IP address used for servers.

? Q11. What is dynamic IP?

✓ Answer:

An IP that changes automatically and is assigned by DHCP.

? Q12. Private vs Public IP?

✓ Answer:

Private → Internal network

Public → Internet communication

? Q13. What is DNS mapping?

✓ Answer:

Converting domain names into IP addresses.

? Q14. What are IP classes?

✓ Answer:

They divide IP addresses into network & host parts.

? Q15. Range of Class A?

✓ Answer:

0 – 127.

? Q16. Range of Class B?

✓ Answer:

128 – 191.

? Q17. Range of Class C?

✓ Answer:

192 – 254.

? Q18. Why Class C usable hosts are 254?

✓ Answer:

Because network ID and broadcast ID are reserved.

? Q19. What is Class D?

✓ Answer:

Used for multicast communication.

? Q20. Why IP is important in cyber security?

✓ Answer:

IP helps in tracking attackers, filtering traffic, and detecting intrusions.

IKM